Transient Low-Order Cues Bootstrap Compositional Representations Across Transformer Scales

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Abstract

We study whether training-distribution geometry can alter formation of a higher-order predictive representation while the target conditional, irreducible noise, marginals, and total predictable information remain fixed. In a finite-field parent-exposed task, two uniform parents determine a modular center and a fixed additive-noise channel determines the target. An exploratory study found that transient full-support low-order cues advanced predictive-state formation. We prospectively froze the architecture, optimizer configuration, objective, evaluation protocol, onset definition, and statistical test before confirmatory training. The confirmatory study used held-out rules, cue permutations, and initializations with three cue doses. Verified onset occurred in 1/8, 6/8, and 8/8 blocks. The one-sided exact blocked Page-style test enumerated 1,679,616 assignments and gave $p = 9.5259869e-06$. All 13 nonzero-cue cells positive at cue removal retained the representation threshold at 6M and 10M. In a preregistered post-confirmatory robustness extension at approximately 1M and 5M core parameters, R2 advanced verified onset in all eight paired blocks. At 1M, onset occurred in 0/4 R0 versus 4/4 R2 cells; at 5M, it occurred in 3/4 versus 4/4. These scale results are descriptive, are excluded from the original v3.6 p-value, and are not a scaling-law fit. Temporary cue-bearing distributions therefore causally changed onset timing or probability within the task and model family under the fixed confirmatory protocol. The experiment does not identify singleton mutual information alone as causal and does not establish a universal Transformer property. In plain terms, a temporary statistical scaffold helped the model discover a rule representation that remained useful after the scaffold was removed.

1 Introduction

The central question is: How does training-distribution geometry affect the accessibility of task-sufficient representations when the final task information is fixed? A representation may be statistically available yet difficult to discover because useful optimization directions are weak, cancel across examples, or become accessible only after other features organize. We therefore intervene on early training-distribution geometry while holding the target conditional and information budget fixed.

Project Kepler uses a finite-field mechanism assay motivated by, but distinct from, its recurrent production task. The parent-exposed design intentionally separates (1) identifying relevant variables, which the interface controls by presenting both parents directly, from (2) composing discovered variables into a predictive representation, which remains the learning problem. The present paper does not claim that the recurrent task has been solved.

Version 3.5 is exploratory discovery. Before confirmatory training, all outcome definitions and analysis choices were prospectively frozen. Version 3.6 is a prospectively frozen confirmatory replication using held-out rules, cue permutations, and initializations; discovery data are excluded from its primary p-value. After that test was complete, we separately preregistered a minimal 1M/5M robustness extension. It reuses the frozen outcomes but does not enter the v3.6 inference. The aim is neither a scaling law nor a universal intelligence

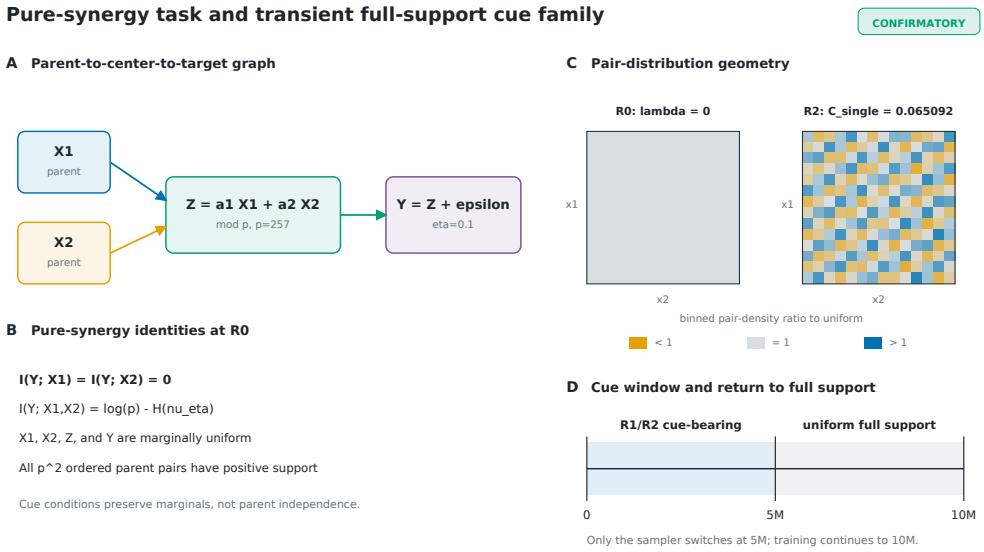


Figure 1: Controlled task, pure-synergy identities, R0-versus-R2 pair-distribution geometry, and transient cue schedule. Binned pair-density ratios and all displayed values are generated from frozen distribution JSON.

principle; it is a bounded causal test plus a post-confirmatory scale check within one task and model family. Figure 1 summarizes the construction and cue schedule.

2 Related Work

Grokking and delayed generalization. Modular-arithmetic grokking studies established that small neural networks can generalize long after achieving near-perfect training performance under a fixed training distribution Power et al. (2022). Follow-up work connected delayed generalization to structured representation learning and its training dynamics Liu et al. (2022). This paper does not offer a universal explanation of grokking. It instead asks whether the geometry of the early training distribution, with the final conditional and information budget held fixed, changes the accessibility or timing of a delayed representational transition.

Mechanistic interpretability of modular arithmetic. Mechanistic analyses of modular addition have recovered Fourier-like internal representations and circuits implementing trigonometric structure Nanda et al. (2023), while other architectures and initializations can learn qualitatively different modular-addition algorithms Zhong et al. (2023). Here, fixed-Fourier probes are diagnostics alongside linear and fixed-capacity MLP probes. Probe accuracy establishes extractability under the specified probe, not causal use by the model Hewitt & Liang (2019); Belinkov (2022). We do not assume that a specific Fourier circuit must be learned, and the causal result concerns the timing or probability of task-sufficient representation formation rather than identification of a unique circuit.

Curriculum learning. Classic curricula order examples by difficulty Bengio et al. (2009); Hacohen & Weinshall (2019); self-paced methods adapt selection to the learner Jiang et al. (2015); and continuation methods smooth or deform an optimization problem Mobahi (2016). The present intervention is a distributional curriculum whose distinctive control is preservation of the target conditional, Bayes entropy, total predictable information, and full pair support. It changes the geometry of a full-support sampling distribution during an early

window. We call this narrower intervention distributional bootstrapping of representation formation.

Optimization path dependence. Learning dynamics can show long plateaus, rapid transitions, and strong dependence on initialization even in analytically tractable deep linear networks Saxe et al. (2014). This motivates studying training paths, but it does not by itself identify the mechanism in the present Transformer. Our gradient statistics are therefore treated as descriptive correlates rather than causal mediators.

Partial Information Decomposition. Partial Information Decomposition separates redundant, unique, and synergistic contributions in multivariate information Williams & Beer (2010). Multiple operationally motivated decompositions exist Bertschinger et al. (2014), so synergy language depends on the adopted definition. The full-support base task is a pure-synergy construction only in the limited operational sense needed here: singleton predictive information is zero while joint predictive information is nonzero. We do not attempt a full PID decomposition of the cue distributions or learned representations.

3 Controlled finite-field task

The field size is $p = 257$. The predictive center is $z = a_1x_1+a_2x_2 \bmod p$, and $y = z+\epsilon \bmod p$ with $P(\epsilon = 0) = 1-\eta$, $P(\epsilon = r) = \eta/(p-1)$ for $r \neq 0$, and $\eta = 0.1$. Thus $H(Y \mid X_1, X_2) = 0.879601$ nats, and the target conditional is unchanged across conditions. The primary metric is

$$R = \frac{\mathbb{E}_h D_{\mathrm{KL}}(q_h \| p_\theta)}{\log p - H(\nu_\eta)}.$$

A uniform predictor has $R = 1$ and the oracle has $R = 0$. Objective, evaluation, bootstrap, and probe details appear in Supplementary Methods.

4 Pure-synergy information structure

Under uniform full support, X_1, X_2, Z, Y are uniform, $I(Y; X_1) = I(Y; X_2) = 0$, and $I(Y; X_1, X_2) = 4.66948$ nats. The scalar cue index orders selected constructed distributions; it is not a unique-information percentage.

5 Full-support cue-family construction

For $z = a_1x_1 + a_2x_2 \bmod p$, the prospectively frozen cue family was

$$P_{\lambda,m}(x_1, x_2) = \frac{1-\lambda}{p^2} + \frac{\lambda}{mp}\mathbf{1}[(z - cx_1) \bmod p \in B_m],$$

where B_m is a frozen size-m subset induced by the cue permutation and c is its frozen cue coefficient. Cue dose was summarized by

$$C_{\mathrm{single}} = \frac{I(Y; X_1) + I(Y; X_2)}{I(Y; X_1, X_2)}.$$

At $\lambda = 0$, the distribution is uniform full support. For every registered nonzero dose, all pair probabilities remain positive and both parent marginals remain uniform, although X_1 and X_2 need not remain jointly independent. Z and Y remain uniform; the target conditional, Bayes entropy, and joint predictable information remain fixed. C_{single} is an operational cue index, not a PID unique-information fraction, and it does not isolate singleton mutual information from other distributional geometry. Figure 1 compares R0 with one cue-bearing distribution. Table 1 gives a representative R0/R1/R2 summary. The complete registered audit is included as machine-readable supplementary material, together with the derivation and deterministic root-solving protocol.

Table 1: Representative confirmatory R0/R1/R2 distributions. H cond. is H(Y|X1,X2); complete registered audits appear in the Supplement.

Condition	C single	Min pair P	Max marg. err.	H cond. (nats)	Joint-I err.	Full support
R0	7.608e-16	1.514e-05	9.107e-18	0.879601	1.066e-14	yes
R1	0.016	1.237e-05	1.735e-18	0.879601	0	yes
R2	0.065092	9.379e-06	1.301e-18	0.879601	0	yes

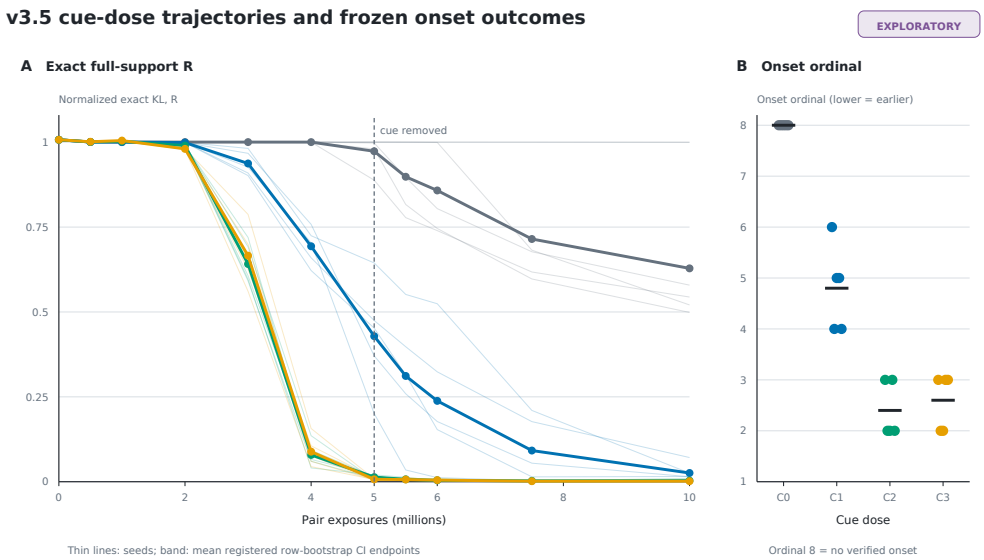


Figure 2: Exploratory v3.5 R trajectories and onset outcomes. Bands are means of registered row-bootstrap endpoints, not new group-level intervals.

6 Exploratory discovery experiment

Version 3.5 used rule $(45, 36)$, five paired seeds, four ordered $m = 64$ doses, and two matched-index geometry controls. Verified onset occurred in 0/5 C0, 5/5 C1, 5/5 C2, and 5/5 C3 seeds. The exploratory exact test gave $p = 1.6075103e - 05$. These observations selected frozen confirmatory conditions and definitions; they are not confirmatory data.

Table 2: Exploratory onset and establishment summary; complete checkpoint coding appears in the Supplement.

Condition	Onset by 5M	Onset by 10M	Established by 10M	Mean R, 5M	Mean R, 10M
C0	0	0	0	0.973341	0.628411
C1	2	5	5	0.428643	0.025721
C2	5	5	5	0.013564	0.003252
C3	5	5	5	0.007108	0.00162

Figure 2 and Table 2 report the discovery trajectories and compact frozen outcomes. Complete coding is included in the machine-readable supplement.

7 Prospectively frozen confirmatory replication

Version 3.6 was a prospectively frozen confirmatory replication using held-out rules, cue permutations, and initializations. Crossing two held-out rules, two held-out cue permuta-

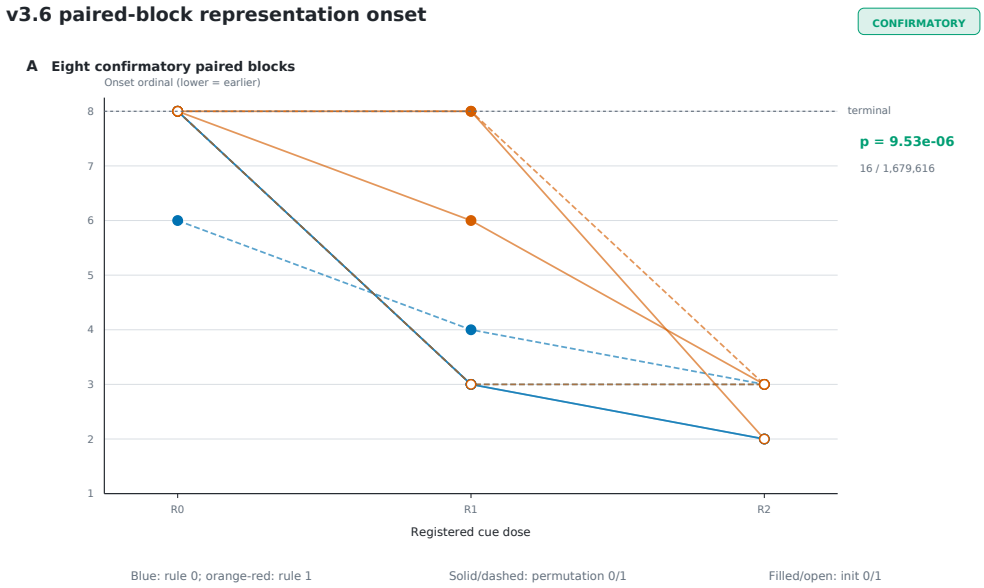


Figure 3: Paired terminal-coded onset ordinals for all eight confirmatory blocks.

tions, and two held-out initialization seeds yielded eight paired blocks and 24 trajectories. R0, R1, and R2 shared initial weights and all non-condition random fields within each block. The primary inference unit was the eight rule × permutation × initialization blocks. The 24 trajectories were paired condition observations—three per block—not 24 independent inferential units. Verified onset occurred in 1/8, 6/8, and 8/8 blocks. The observed Page-style statistic was 110.0. Under the prospectively frozen sharp null, condition labels were exchangeable within each block. Exactly 16 of 1,679,616 assignments were at least as large; the exact upper-tail count included the observed assignment. This gave one-sided $p = 9.5259869e - 06$.

Table 3: Confirmatory outcomes across eight paired inference blocks; complete coding appears in the Supplement.

Condition	Onset by 5M	Onset by 10M	Established by 10M	Mean R, 5M	Mean R, 10M
R0	0	1	2	1.000239	0.432743
R1	5	6	6	0.303076	0.110766
R2	8	8	8	0.04371	0.005181

Table 4: Primary confirmatory exact test. Stratified tests are secondary, nominal, and not multiplicity-adjusted; full results appear in the Supplement.

Analysis	Blocks	Assignments	Statistic	Upper tail	One-sided p
primary	8	1679616	110	16	9.526e-06

Figures 3 and 4 show the paired onset ordinals and exact null. Tables 3, 4, and 5 report compact confirmatory outcomes, the primary exact test, and mean paired differences. Complete fields and the transition-coding audit are in the machine-readable supplement. Rule- and permutation-stratified tests are secondary, nominal, and not multiplicity-adjusted.

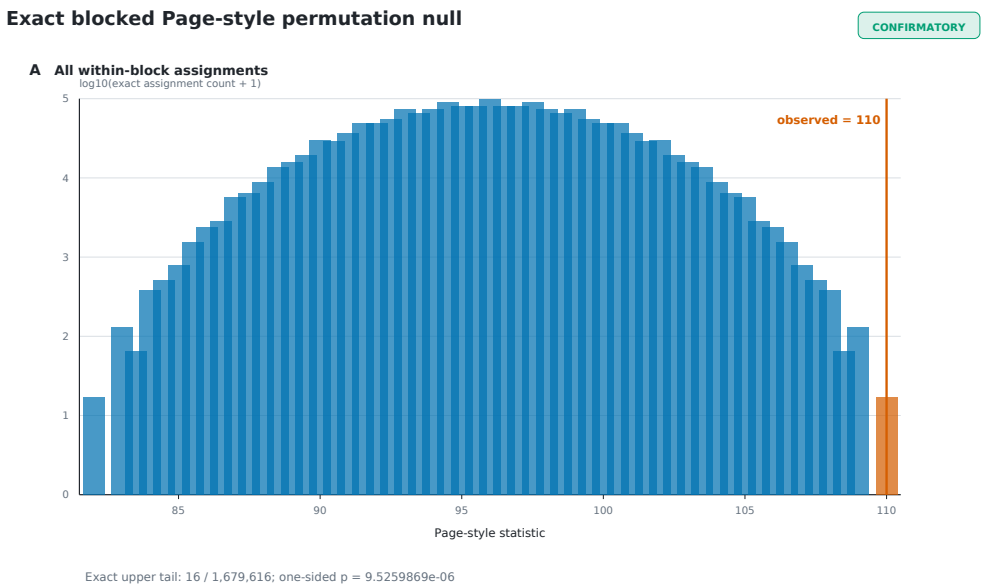


Figure 4: Exact v3.6 permutation null. Version 3.5 is excluded.

Table 5: Mean paired differences, higher registered dose minus lower registered dose; medians and audit fields appear in the Supplement.

Comparison	Blocks	Mean delta R, 5M	Mean delta R, 10M	Mean onset delta
R1_minus_R0	8	-0.697162	-0.321977	-3
R2_minus_R0	8	-0.956528	-0.427562	-5.125
R2_minus_R1	8	-0.259366	-0.105585	-2.125

8 Representation persistence after cue removal

At 5M only the sampler changed for nonzero-cue runs. Zero-step checks verified unchanged outputs and model, optimizer, scheduler, exposure, and RNG state. All 13 threshold-positive R1/R2 cells remained positive at 6M and 10M, supporting learning-path preconditioning rather than a permanently required shortcut.

Table 6: Confirmatory retention after return to uniform full support; complete exploratory and confirmatory records appear in the Supplement.

Condition	Positive at 5M	Retained at 6M	Retained at 10M
R1	5	5	5
R2	8	8	8

Figure 5 and Table 6 report the persistence result. The complete exploratory and confirmatory switch audit is included in the machine-readable supplement.

9 Cross-scale context and post-confirmatory robustness

The original v3.6 confirmatory experiment used approximately 2M core parameters. After completing it, we preregistered a minimal extension at approximately 1M and 5M core parameters. Width was the only scaled architectural quantity; depth, heads, input interface, $p = 257$, $\eta = 0.1$, optimizer parameterization, exact soft-label objective, and absolute-exposure schedule were unchanged. Two frozen rules and two paired initialization seeds

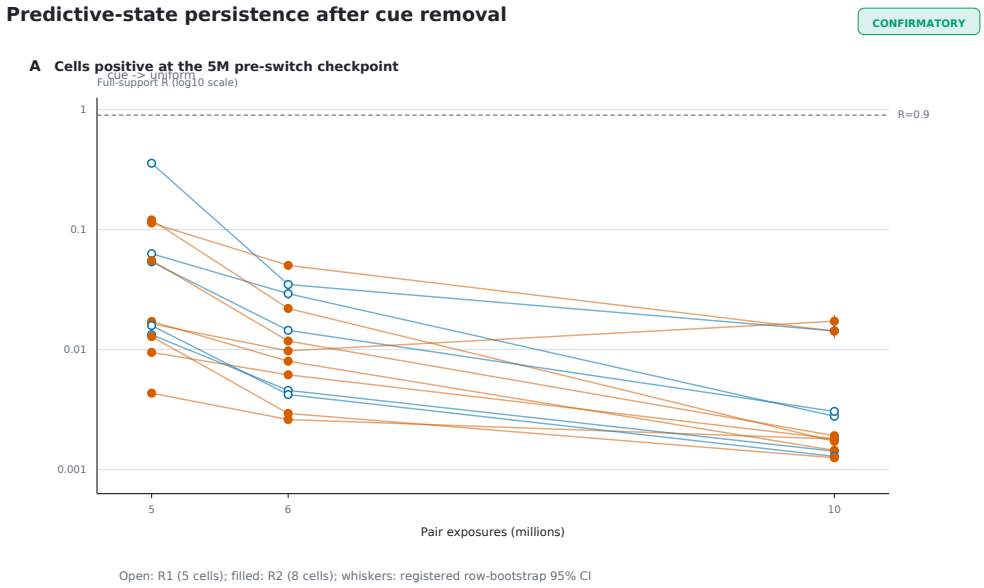


Figure 5: Persistence after return to uniform full support. Whiskers are registered first-parent-row bootstrap intervals.

Table 7: R0/R2 onset across the original confirmatory scale and the post-confirmatory robustness scales. "Earlier" compares the registered terminal-coded onset ordinal within each paired block. Rows are not pooled: no combined p-value or scaling-law fit is computed.

Scale	Status	Blocks	R0 onset	R2 onset	R2 earlier	Mean Δ
∼1M	Post-conf. robustness	4	0/4	4/4	4/4	-5.25
∼2M	Confirmatory v3.6	8	1/8	8/8	8/8	-5.125
∼5M	Post-conf. robustness	4	3/4	4/4	4/4	-3.75

were crossed with R0 and R2, giving four paired blocks per scale and 16 trajectories. R0/R2 shared initialization and non-condition randomness within every block.

A preregistered engineering/learnability pilot first ran one pair per scale. Both scales passed memory, throughput, checkpoint-resume, non-saturation, and identifiability gates; a favorable cue effect was not required. All 16 runs then used cue-bearing sampling through 5M and uniform full support from 5M to 10M. All zero-step switch checks and all 176 checkpoint sidecar checks passed.

Table 7 places the R0/R2 subset of the original 2M confirmatory result beside the new robustness scales. R2 advanced the registered terminal-coded onset ordinal in 8/8 confirmatory 2M blocks and in 4/4 post-confirmatory blocks at each of 1M and 5M. Verified onset occurred in 1/8 R0 versus 8/8 R2 trajectories at 2M, 0/4 versus 4/4 at 1M, and 3/4 versus 4/4 at 5M. All new R2 trajectories retained the onset threshold at 6M and 10M after cue removal. Every paired block and its frozen metrics appear in Tables 8–9. The 1M/5M extension is post-confirmatory descriptive evidence: it is excluded from the original v3.6 p-value. The three display rows are not pooled, no new combined p-value is computed or implied, and three selected widths do not constitute a scaling-law fit.

10 Rule and cue-geometry heterogeneity

R1 onset occurred in 4/4 rule-0 and 2/4 rule-1 blocks, while R2 onset occurred in 4/4 and 4/4. The ordered effect spans both rules but the mid-dose response is rule-dependent.

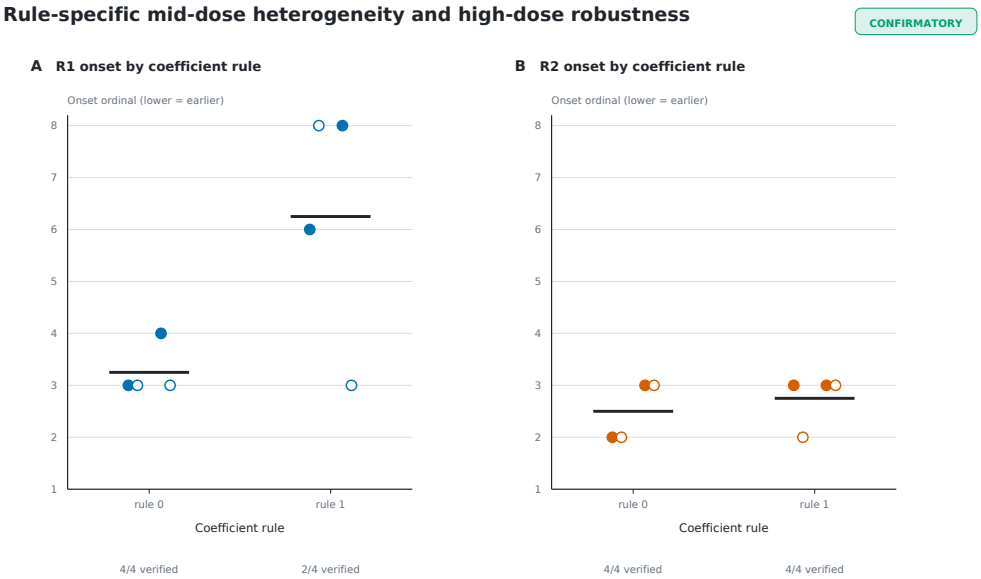


Figure 6: Rule-specific R1 heterogeneity and R2 robustness.

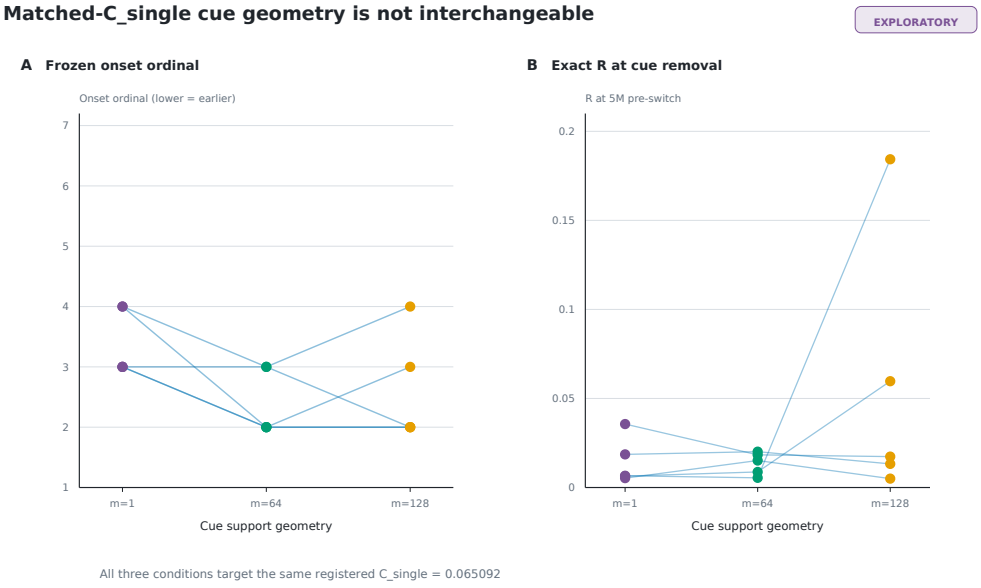


Figure 7: Exploratory matched-index geometry comparison.

In exploratory matched-index controls, concentrated and diffuse geometries differed from $m = 64$, so equal cue-index values are not interchangeable.

Figures 6 and 7 separate the confirmatory rule comparison from the exploratory geometry comparison.

11 Discussion

Temporary cue-bearing full-support distributions changed whether or when the Transformer trained under the prospectively frozen protocol formed the modular predictive center. The target conditional, marginals, irreducible noise, total pairwise predictable information, architecture, optimizer, and objective were held fixed. Persistence after cue removal indicates that the early distribution altered the learning path rather than supplying a permanently required feature. Heterogeneity across rules and geometries rejects an explanation based on the scalar cue index alone. The post-confirmatory 1M/5M extension preserved the paired direction in all eight new blocks, while the differing R0 onset rates show that scale changes baseline learnability. This robustness result does not extend the original confirmatory inference or justify a scaling law.

11.1 Possible mechanism: symmetry breaking and optimization accessibility

Pure synergy removes singleton predictive signal from the population distribution. In the tested parameterization, this condition was associated with weak and poorly aligned early gradients. Cue-bearing distributions introduced lower-order statistical asymmetries that may have changed early gradient alignment and optimization accessibility. One possible consequence is an earlier path toward composing both parents; after the task-sufficient representation forms, the cue is no longer required. This is an experiment-bounded interpretation, not a theorem that gradients generally vanish under pure synergy.

Three observations are consistent with this account: onset was ordered by registered cue dose, threshold-positive representations persisted after cue removal, and the frozen gradient audits showed changes in gradient coherence around observed transitions. This is mechanistic interpretation, not a causal mediation result. Gradient statistics are correlates of transition dynamics, not identified causal mediators. The data do not establish a universal loss-landscape mechanism or a unique circuit.

11.2 Practical implication: distributional bootstrapping

The result suggests a possible design principle: temporary training distributions can be used to improve access to otherwise hard-to-reach compositional representations when the final task distribution contains higher-order dependencies. This is a hypothesis for bounded, auditable interventions, not a claim that all AI training should use low-order cues.

12 Limitations

First, model scale is sampled only at approximately 1M, 2M, and 5M core parameters, with just four post-confirmatory pairs at each new scale; this is not a scaling-law design. Second, the parent-exposed interface intentionally controls variable identification; recurrent context discovery and variable selection remain future work. Third, exact soft-label training isolates representation formation from sampled-label variance; sampled-token training remains future work. Fourth, finite-field modular arithmetic is a controlled model organism rather than a proxy for the full range of natural-data learning. Fifth, the experiments identify a distribution intervention but do not establish a universal loss-landscape mechanism.

The study also uses one field size, one noise level, a finite checkpoint grid, and eight confirmatory blocks. The exact test is valid for the frozen randomization structure but does not make the tested rules a population sample of all modular tasks. The 0–5M cue window coincided with the prospectively frozen 5M learning-rate warmup, and cue timing was not crossed with optimizer phase. Cue conditions preserved both parent marginals but changed their joint distribution, so the intervention is not a marginal-frequency manipulation alone. Dependence beyond the tested widths, field size, noise level, input interface, and exact soft labels remains open. The result is not a universal Transformer claim, not a thermodynamic phase transition, and not evidence that the recurrent production task is solved. These results are excluded from the original v3.6 p-value.

Reproducibility statement

The confirmatory and post-confirmatory results and their frozen hashes are recorded in the anonymous supplementary artifact package. Frozen hashes, environment, test definitions, counts, and provenance are indexed in the package. The lightweight verifier recomputes distribution audits, transition coding, all 1,679,616 exact v3.6 assignments, scale summaries, figures, and tables without optimizer steps.

Data and code availability

Code, manifests, result JSON, source data, and verification scripts are included in the repository. Checkpoints are referenced by SHA-256 sidecars and are unnecessary for lightweight verification.

AI use statement

Generative AI tools were used to discuss and refine hypotheses and experimental methodology; implement, review, and test research software; inspect registered results; and draft, edit, structure, and format the manuscript and artifacts. No generative model supplied or altered experimental observations or labels; all synthetic task data were produced by audited code. The human authors selected and approved all preregistrations, stopping rules, analyses, interpretations, and claims; they verified AI-assisted code with automated tests and frozen hash audits and reviewed the reported values against machine-readable results. The authors take responsibility for the final content, including text, claims, code, and artifacts produced with the aid of generative AI.

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A Supplementary methods

A.1 Task objective, evaluation, and probes

Exact soft-label cross-entropy was used throughout the parent-exposed experiments: at each observed pair, the target was the analytic noisy conditional rather than one sampled label. Evaluation enumerated all 66,049 ordered parent pairs. Every checkpoint recorded exact conditional KL, normalized R, center accuracy, center probability, predictive entropy, and a 2,000-replicate first-parent-row bootstrap interval. Selected checkpoints included layer-wise linear, fixed-Fourier, and fixed-capacity MLP probes.

A.2 Cue-family construction and audit

Frozen cue permutations and coefficients reweighted the complete pair space while preserving strictly positive probability for every pair. Lambda was solved by deterministic root finding for each prospectively frozen target, not selected from model outcomes. Exhaustive enumeration verified marginal uniformity, conditional invariance, joint information, and full support.

A.3 Model and optimization

The v3.6 confirmatory manifest instantiated the architecture and optimization settings frozen from v3.5. It records 2,045,568 core parameters, width 168, batch size 4,096, exact soft-label cross-entropy, AdamW learning rate 3×10^{-4}, betas $(0.9, 0.95)$, weight decay 0.1, clipping at 1.0, 5M warmup exposures, and an absolute-exposure cosine schedule.

A.4 Transition coding

Probe checkpoints were 2M, 3M, 4M, 5M, 6M, 7.5M, and 10M. Onset required $R < 0.9$ and layer-6 linear accuracy above 0.5, retained at the next checkpoint. Establishment required $R < 0.1$, center accuracy above 0.9, and layer-6 linear accuracy above 0.9.

A.5 Exact test

Within-block terminal onset ordinals were converted to doubled midranks, with earlier onset ranked higher. All six label assignments were combined across eight blocks, giving $6^8 = 1,679,616$ assignments. Under the sharp null, the R0/R1/R2 labels were exchangeable within each block. The one-sided upper tail included the observed assignment.

A.6 Post-confirmatory scale protocol and complete paired outcomes

The separately frozen scale manifest crossed target core-parameter counts 1M and 5M with two v3.6 coefficient rules, two v3.6 initialization seeds, and R0 versus R2. Widths 120

Table 8: Every post-confirmatory paired block. Onset and establishment use the unchanged v3.6 definitions. A dash denotes no verified event by 10M; persistence reports whether the R2 onset threshold was retained at 6M and 10M after cue removal.

Scale	Rule	Seed	R0 onset	R2 onset	Δ	R0/R2 est.	R2 persist.
1M	0	0	–	4M	-5	– / 5M	yes / yes
1M	0	1	–	3M	-6	– / 5M	yes / yes
1M	1	0	–	4M	-5	– / 6M	yes / yes
1M	1	1	–	4M	-5	– / 7.5M	yes / yes
5M	0	0	–	3M	-6	– / 4M	yes / yes
5M	0	1	5M	3M	-2	6M / 3M	yes / yes
5M	1	0	7.5M	3M	-4	10M / 4M	yes / yes
5M	1	1	6M	3M	-3	10M / 4M	yes / yes

Table 9: Registered 10M outcomes in every post-confirmatory paired block. Columns give exact full-support R, exact center accuracy, and the registered layer-6 linear-probe accuracy for R0/R2.

Scale	Rule/seed	R_0	R_2	Center_0	Center_2	Probe_0	Probe_2
1M	0/0	.5982	.0027	.0492	1.0000	.1333	1.0000
1M	0/1	.7063	.0038	.0257	1.0000	.0895	1.0000
1M	1/0	.5785	.0045	.0766	1.0000	.1346	1.0000
1M	1/1	.5781	.0221	.0475	.9832	.1749	.9887
5M	0/0	.4992	.0007	.1154	1.0000	.1718	1.0000
5M	0/1	.0028	.0001	.9987	1.0000	.9995	1.0000
5M	1/0	.0303	.0002	.9795	1.0000	.9968	1.0000
5M	1/1	.0413	.0003	.9632	1.0000	.9676	1.0000

and 264 were selected by the unchanged width-only rule, yielding 1,046,400 and 5,039,232 actual core parameters. Each paired block shared initialization and every non-condition random field. The target conditional, exact soft-label objective, optimizer protocol, 5M switch, and registered probe grid were unchanged. The pilot gate used only engineering and identifiability criteria. The analysis reports all blocks descriptively and does not combine them with the v3.6 randomization test.

A.7 Exploratory v1.3 Hessian audit

Version 1.3 was a post hoc, no-training audit of six frozen checkpoints from one representative v3.6 paired block. It applied exact Hessian-vector products on CPU to a fixed 512-pair full-support sample, using Lanczos Ritz estimates for leading eigenvalues and a fixed-seed Hutchinson trace estimate. The spectral quantities changed sharply around the selected transition checkpoints (Table 10), but this single-block analysis was not preregistered as a mechanistic test. It neither identifies a cue-induced loss-landscape change nor establishes Hessian or gradient quantities as causal mediators. It is included only as exploratory appendix material and does not enter the v3.6 or scale-robustness conclusions.

Table 10: Exploratory v1.3 Hessian audit on six frozen checkpoints from one representative paired block. Values are CPU Lanczos/Hutchinson estimates on one fixed 512-pair evaluation sample and are descriptive signatures, not causal mediators.

Condition	Role (exposure)	$\lambda_{\max}$	Trace	$d_{\text{eff},10}$
R0	pre (6M)	143.06	1,418.77	6.414
R0	transition (7.5M)	6,337.52	64,422.7	5.828
R0	late (10M)	2,596.69	22,164.1	6.354
R2	pre (3M)	3,155.68	75,057.1	8.486
R2	transition (4M)	9,993.67	194,199	8.187
R2	post (5M)	10,634.2	194,973	8.146

A.8 Artifact integrity

The freeze records every tag, manifest and result hash, the environment lock, test definition and result, and artifact counts. Final packaging adds only the manuscript, audit records, and hash registries; no prior scientific artifact is modified.

A.9 Machine-readable audit material

The anonymous code supplement retains every registered audit and provenance field as JSON, CSV, and Markdown. The compact manuscript tables are presentation-only views of those frozen values. Checkpoint binaries are omitted from the submission archive; their hashes and verification outcomes are retained.